An Elementary Proof of the Shoelace Formula

1 Introduction

The **Shoelace Formula** is an elegant method for computing the area of a polygon given the coordinates of its vertices. It tells us that if we have a simple (not self-intersecting) polygon with vertices $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$ listed in counterclockwise order, its area is given by

$$A = \frac{1}{2}\left[\sum_{i=1}^{n}(x_i y_{i+1} - x_{i+1} y_i)\right], \tag{1}$$

where we use the convention that $(x_{n+1}, y_{n+1}) = (x_1, y_1)$. This formula, which was first described in 1769 [1], is called the "shoelace formula" because the pattern of multiplying coordinates resembles the crisscross pattern of shoelaces. We will prove this formula in two steps. First, we establish it for triangles, and then we extend the result to arbitrary simple polygons using mathematical induction. While there are already other proofs for the formula using induction [2], as far as we know the following proof is not publically available elsewhere.

2 The Shoelace Formula for Triangles

Theorem 1. *Let $A = (x_1, y_1)$, $B = (x_2, y_2)$, and $C = (x_3, y_3)$ be the vertices of a triangle listed in counterclockwise order. Then the area of triangle ABC is given by:*

$$\mathit{Area}(ABC) = \frac{1}{2}\left((x_1 y_2 - x_2 y_1) + (x_2 y_3 - x_3 y_2) + (x_3 y_1 - x_1 y_3)\right) \tag{2}$$

Proof. We begin by shifting the coordinate system so that vertex A is placed at the origin as shown in Figure 1. Define the shifted triangle as $A'B'C'$ as

$$\begin{aligned} A' &= (0, 0) \\ B' &= B - A = (x_2 - x_1, y_2 - y_1) \\ C' &= C - A = (x_3 - x_1, y_3 - y_1). \end{aligned}$$

Note that simply shifting the triangle does not change its area, so $\text{Area}(ABC) = \text{Area}(A'B'C')$. Let us denote the position vectors of B' and C' as $\vec{b} = (b_x, b_y)$ and $\vec{c} = (c_x, c_y)$, respectively. We also add a new point P at $B' + C' = (b_x + c_x, b_y + c_y)$ so that $A'B'PC'$ forms a parallelogram.

Then we use the standard area formula for the parallelogram $A'B'PC'$ in terms of its two sides b and c (which is also exactly the definition of the cross-product)

$$\text{Area}(A'B'PC') = b_x c_y - b_y c_x.$$

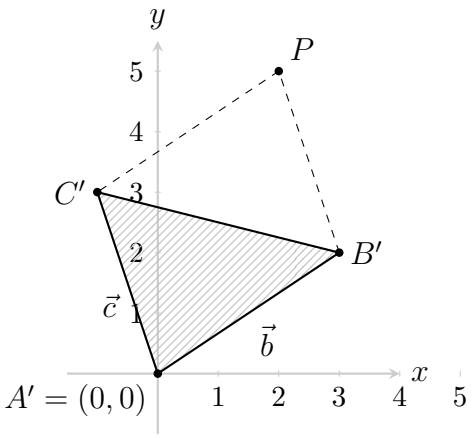


Figure 1: The shifted triangle $A'B'C'$ with A' at the origin.

In case you are not familiar with this formula, there are also nice visualisations leading to a short proof [3]. Since the triangle $A'B'C'$ occupies exactly half of the parallelogram, we have

$$\text{Area}(A'B'C') = \frac{1}{2}\left(b_x c_y - b_y c_x\right).$$

Now we substitute back the original coordinates. Recall that

$$\begin{aligned} b_x &= x_2 - x_1, \quad & b_y &= y_2 - y_1 \\ c_x &= x_3 - x_1, \quad & c_y &= y_3 - y_1. \end{aligned}$$

Therefore

$$\begin{aligned} \text{Area}(ABC) &= \frac{1}{2}\left(b_x c_y - b_y c_x\right) \\ &= \frac{1}{2}\left((x_2 - x_1)(y_3 - y_1) - (y_2 - y_1)(x_3 - x_1)\right) \\ &= \frac{1}{2}\left(x_2 y_3 - x_2 y_1 - x_1 y_3 + x_1 y_1 - y_2 x_3 + y_2 x_1 + y_1 x_3 - x_1 y_1\right) \\ &= \frac{1}{2}\left(x_2 y_3 - x_2 y_1 - x_1 y_3 - x_3 y_2 + x_1 y_2 + x_3 y_1\right) \\ &= \frac{1}{2}\left((x_1 y_2 - x_2 y_1) + (x_2 y_3 - x_3 y_2) + (x_3 y_1 - x_1 y_3)\right). \end{aligned}$$

This completes the proof for triangles. □

3 The Shoelace Formula for Simple Polygons

Theorem 2. *Let P be a simple polygon with n vertices $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$ listed in counterclockwise order. Then the area of P is*

$$\textit{Area}(P) = \frac{1}{2}\left(\sum_{i=1}^{n}(x_i y_{i+1} - x_{i+1} y_i)\right),$$

where $(x_{n+1}, y_{n+1}) = (x_1, y_1)$.

Proof. We proceed by induction on the number of vertices n.

Base Case ($n = 3$): This is precisely Theorem 1, which we have already established. The shoelace formula holds for triangles.

Inductive Step: Assume that the shoelace formula holds for all simple polygons with n vertices. We will show that it also holds for simple polygons with $n + 1$ vertices.

For notational convenience, we denote the shoelace sum of a polygon with vertices $P_1, P_2, \ldots, P_k$ by $S(P_1P_2 \cdots P_k)$.

Consider a simple polygon with vertices $P_1, P_2, \ldots, P_n$ listed in counterclockwise order. We will add a new vertex P_{n+1} to form a polygon with $n + 1$ vertices by replacing edge P_nP_1 with edges P_nP_{n+1} and $P_{n+1}P_1$. There are three cases to consider:

Case 1: Point P_{n+1} lies outside the polygon $P_1P_2 \cdots P_n$.

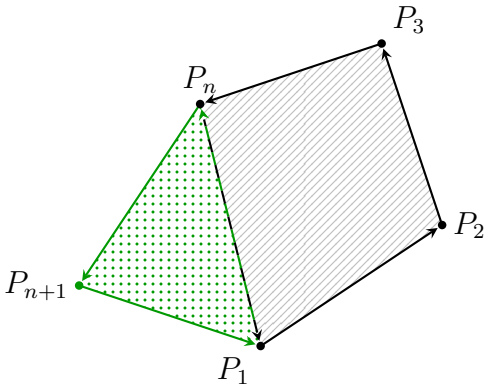


Figure 2: Point P_{n+1} lies outside the n-vertex polygon. (Case 1)

In this case, the $(n + 1)$-vertex polygon is $P_1P_2 \cdots P_nP_{n+1}$ as shown in Figure 2.

When we traverse the boundary of the n-vertex polygon in the counterclockwise direction, we include the edge from P_n to P_1. When we traverse the triangle $P_1P_nP_{n+1}$ in the counterclockwise direction, we include the edge from P_1 to P_n. These two edges are opposite in direction, so their contributions to the shoelace sum cancel out:

$$\begin{aligned} &\text{Contribution from } P_n \to P_1: \quad x_ny_1 - x_1y_n \\ &\text{Contribution from } P_1 \to P_n: \quad x_1y_n - x_ny_1 \end{aligned}$$

Therefore, the shoelace sum for the $(n + 1)$-vertex polygon equals the sum of the shoelace sums for the n-vertex polygon and the triangle, minus the contributions from the shared edge. Since these contributions cancel, we have

$$S(P_1P_2 \cdots P_nP_{n+1}) = S(P_1P_2 \cdots P_n) + S(P_1P_nP_{n+1}).$$

By the inductive hypothesis, the shoelace formula holds for the n-vertex polygon $P_1P_2 \cdots P_n$ and for the triangle $P_1P_nP_{n+1}$.

$$= \text{Area}(P_1P_2 \cdots P_n) + \text{Area}(P_1P_nP_{n+1}) = \text{Area}(P_1P_2 \cdots P_nP_{n+1})$$

This establishes the shoelace formula for Case 1.

Case 2: Point P_{n+1} lies inside the polygon $P_1P_2 \cdots P_n$.

Again the $(n+1)$-vertex polygon is $P_1P_2 \cdots P_nP_{n+1}$, but now the point P_{n+1} is in the interior of the original n-vertex polygon $P_1P_2 \cdots P_n$ as shown in Figure 3.

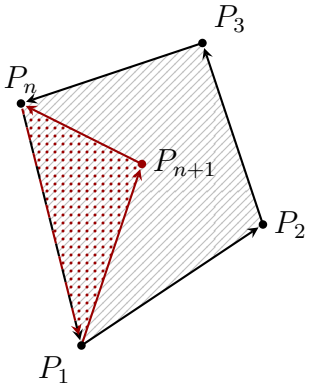


Figure 3: Point P_{n+1} lies inside the n-vertex polygon. (Case 2)

Like in case 1, the edge (P_1, P_n) is included twice, but this time both occurrences point in the same direction and so do not cancel each other out. However, observe that if we invert the order of the vertices, each term of the shoelace sum is multiplied by -1:

$$x_i y_{i+1} - x_{i+1} y_i = (-1)\,(x_{i+1} y_i - x_i y_{i+1})$$

Thus, when we invert the order the sum S is also simply multiplied by -1 and we get

$$S(P_1 P_{n+1} P_n) = -S(P_1 P_n P_{n+1}).$$

Now we have again one edge from P_1 to P_n and another from P_n to P_1, cancelling each other out. So we can apply the same argument as last time:

$$\begin{aligned} S(P_1 P_2 \cdots P_n P_{n+1}) &= S(P_1 P_2 \cdots P_n) + S(P_1 P_n P_{n+1}) \\ &= S(P_1 P_2 \cdots P_n) - S(P_1 P_{n+1} P_n) \\ &= \text{Area}(P_1 P_2 \cdots P_n) - \text{Area}(P_1 P_{n+1} P_n) \\ &= \text{Area}(P_1 P_2 \cdots P_n P_{n+1}) \end{aligned}$$

Self-intersecting polygons

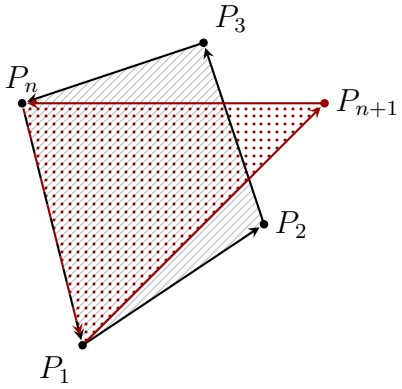


Figure 4: Point P_{n+1} positioned such that edges intersect.

Finally it should be mentioned that adding the P_{n+1} could cause the polygon to self-intersect as shown in Figure 4. But then the polygon is not simple anymore and we had

specifically excluded such polygons in theorem 2. In pratice, the Shoelace formula is often actually used to give such polygons a definable area.

By considering Cases 1 and 2, we have shown that the shoelace formula holds for all simple polygons with $n + 1$ vertices, assuming it holds for polygons with n vertices. By the principle of mathematical induction, the shoelace formula holds for all simple polygons. □

References

[1] Albrecht Ludwig Friedrich Meister. *Generalia de genesi figurarum planarum et inde pendentibus earum affectionibus. Novi Commentarii Societatis Regiae Scientiarum Gottingensis*, 1:144, 1769. In Latin.

[2] lhf Proving the Shoelace Formula with Elementary Calculus. Math StackExchange. `https://math.stackexchange.com/q/809839`.

[3] littleO. Why determinant of a 2 by 2 matrix is the area of a parallelogram? Math StackExchange. `https://math.stackexchange.com/q/4318391`.